Methodology

Carbon Monoxide (CO) Analysis Using Sentinel-5P and Google Earth Engine

1. Objective

The objective of this project is to analyze the spatial distribution of atmospheric Carbon Monoxide (CO) concentration over the selected Area of Interest (AOI) using Sentinel-5P OFFL Level-3 data in Google Earth Engine (GEE). The analysis produces an annual mean CO concentration map along with descriptive statistics for the study area.

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2. Study Area

The analysis was conducted over a user-defined Area of Interest (AOI). All satellite images were clipped to the AOI before visualization and statistical analysis.

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3. Dataset

Satellite: Sentinel-5P

Product: OFFL Level-3 Carbon Monoxide (CO)

Dataset ID:

COPERNICUS/S5P/OFFL/L3_CO

Band Used:

CO_column_number_density

Spatial Resolution: Approximately 1 km

Platform: Google Earth Engine

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4. Study Period

Start Date: **1 January 2025**

End Date: **31 December 2025**

An annual mean composite was generated from all available observations during this period.

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5. Methodology

Step 1: Load the Area of Interest

The Area of Interest (AOI) was imported into Google Earth Engine and displayed on the map.

Step 2: Load Sentinel-5P CO Dataset

The Sentinel-5P OFFL Level-3 Carbon Monoxide Image Collection was loaded using the Google Earth Engine data catalog.

Step 3: Filter Images

The image collection was filtered according to:

- Study area (AOI)
- Study period (2025)

Step 4: Generate Annual Mean Composite

The mean() reducer was applied to create a single annual average CO concentration image representing the study period.

Step 5: Clip to Study Area

The annual mean image was clipped to the AOI to limit the analysis to the selected study region.

Step 6: Visualization

The CO concentration map was visualized using a color palette ranging from low to high concentration values.

Step 7: Statistical Analysis

Descriptive statistics were calculated using the reduceRegion() function.

The following statistics were derived:

- Mean
- Minimum
- Maximum

(Optional extensions include Median and Standard Deviation.)

Step 8: Export Results

The following outputs were exported:

- Annual Mean CO Map (GeoTIFF)
- Descriptive Statistics (CSV)

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6. Software and Tools

- Google Earth Engine
- JavaScript API
- QGIS / ArcGIS Pro (optional for visualization)

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7. Output Products

- Annual Mean Carbon Monoxide Map (.tif)
- Carbon Monoxide Statistics (.csv)

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8. Workflow Summary

1. Import AOI
2. Load Sentinel-5P CO dataset
3. Filter by date and location
4. Compute annual mean
5. Clip to AOI
6. Visualize CO concentration
7. Calculate descriptive statistics
8. Export GeoTIFF and CSV outputs

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9. Notes

- CO concentration is represented as column number density.
- Annual averaging reduces the effect of short-term atmospheric variability and highlights long-term spatial patterns.
- The methodology can be adapted for monthly, seasonal, or multi-year analyses by modifying the date range.